

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM SIX MOCK EXAMINATION

CHEMISTRY-3A
(ACTUAL PRACTICAL)

GEPAM science hub

WhatsApp 0748978398

February 2025 A.M

INSTRUCTIONS

1. This paper consists of **three (3)** questions. Answer **all** questions.
2. Question number **one (1)** carries **twenty (20)** marks and the other **two (2)**, carry **fifteen (15)** marks each.
3. Qualitative Analysis Guide (QAG) sheet authorized by NECTA may be used.
4. Mathematical tables and non – programmable calculators may be used.
5. All writing must be in **blue or black ink except** drawing which must be in pencil.
6. Communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination number** every page of your answer booklet(s).
8. You may use the following atomic masses:
 $H = 1, C = 12, N = 14, O = 16, S = 32, Na = 23, Cl = 35.5, K = 39, Mn = 55, Fe = 56, Mg = 24.$

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This paper consists of 3 printed pages

1. You are provided with the following:

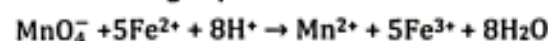
A₁: A solution of sodium oxalate, Na₂C₂O₄, made by dissolving 3.35 g of the salt in a 0.5 dm³ of a solution.
A₂: A solution of potassium permanganate KMnO₄;
A₃: A solution of hydrated iron (II) ammonium sulphate, FeSO₄(NH₄)₂SO₄·XH₂O made by dissolving 33.3 g of the salt in a distilled water to form a 1dm³ of an aqueous solution.
A₄: Dilute sulphuric acid;

Thermometer.

Theory

Standardization of A₂ solution is done by titrating it against the standard solution of A₁ in acidic medium. The resulting reaction equation is as follows:

$2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 10\text{CO}_2$. The standardized KMnO₄ is then titrated against A₃, whose number of molecules of water of crystallization can then be calculated. The resulting equation is as follows:



Procedure

Part I

- Measure 20 cm³ of solution A₁ into a titration flask and then add 10 cm³ of solution A₄.
- Heat the contents near to boiling (about 80°C).
- Titrate this hot mixture against solution A₂ from the burette until there is a colour change.
- Repeat the procedures (i) to (iii) to obtain three more readings and record your titration results in a tabular form.

Summary

_____ cm³ of a solution A₁ required _____ cm³ of a solution A₂ for complete reaction.

Part II

- Measure 20 cm³ of a solution A₃ into a titration flask and then add 10 cm³ of a solution A₄.
- Titrate the reaction mixture against the solution A₂ from the burette until there is a colour change.
- Repeat the procedures (i) to (ii) to obtain three more readings and record your results in tabular form.

Questions

- Calculate the:
 - Molarity of potassium permanganate.
 - Concentration of potassium permanganate in gdm⁻³.
 - Molarity of Iron (II) salt.
 - Concentration of anhydrous iron (II) salt in gdm⁻³.
- Find the value of X in the formula FeSO₄(NH₄)₂SO₄·XH₂O.

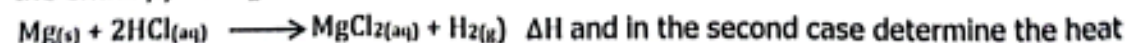
2. You are provided with the following:

U: A solution of 2 M Hydrochloric acid;
V: 0.2 g of magnesium ribbon;
W: 1 g of magnesium carbonate;

Thermometer.

Theory

You are required to determine the heat of formation of MgCO₃. In the first case determine the enthalpy change for the reaction



change for the reaction



The heat evolved can be calculated from the following formulae:

$$\text{Heat evolved } (\Delta H) = -c\rho V\Delta T$$

Where c = Specific heat capacity of water

ρ = density of solution

V = Volume of solution.

ΔT = Temperature change.

$$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$$

$$\Delta H_f = (-A) + (-394) + (-286) + B$$

Procedures

Part I

- Measure 50 cm³ of **U** into a 100 cm³ beaker or conical flask.
- Determine the initial temperature T_i .
- Add 0.2 g of **V** in (i) above. Swirl the mixture and record the final temperature T_f using a thermometer.

Part II

- Measure 50 cm³ of **U** into a 100 cm³ glass beaker or conical flask.
- Determine the initial temperature T_i .
- Add 1 g of **W** in (i) above. Swirl the mixture and record the final temperature T_f using a thermometer.

Questions

- Calculate the heat evolved during the reaction in part I and II, given that, the specific heat capacity of the solution = 4.2 Jg⁻¹°C⁻¹ and density of the solution = 1 gcm⁻³. Neglect the heat absorbed by the container and assume no change in the volume of the solution.
- Calculate the enthalpy of formation of magnesium carbonate (MgCO₃), given that enthalpy of formation of CO₂ = -394 kJmol⁻¹ and enthalpy of formation of H₂O = -286 kJmol⁻¹

3. You are provided with sample **M** containing **two cations** and **one anion**. Carry out experiments described in table 1. Record carefully your observations make appropriate inference and finally the cation and anion present in sample **M**.

No.	Experiment	Observation	Inference
(a)	Observe the appearance of sample M		
(b)	Perform flame test for sample M .		
(c)	Heat a little sample M in a dry test tube.		
(d)	Add concentrated sulphuric acid to the small portion of the sample.		
(e)	To the small portion of the prepared solution, add dilute sodium hydroxide.		
(f)	To the small portion of the solution, add dilute HCl followed by hydrogen sulphide. Filter the precipitates to obtain filtrate and residue then proceed as follows;		
	(i) To the filtrate, add potassium ferrocyanide (III).		
	(ii) Dissolve the residue in aqua regia and then add excess 50% ammonia solution.		
(e)	To the small portion of the solution, add dilute nitric acid followed by silver nitrate.		

Conclusion

- The cations in sample **M** are _____ and _____
- The anion in sample **M** is _____
- The molecular formula **M** is _____

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MARKING SCHEME.

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Part I

Table of results.

Experiment	Pilot	1	2	3
Final volume (cm ³)	20.50	20.20	20.00	20.10
Initial volume (cm ³)	0.00	0.00	0.00	0.00
Volume used (cm ³)	20.50	20.20	20.00	20.10

Filling the table = 01 Mark
 Decimal places = 01 Mark
 Accurate volume = 01 Mark.
 Total = (03) Marks

$$\begin{aligned}
 \text{Average volume used} &= \frac{T_1 + T_2 + T_3}{3} \\
 &= \frac{(20.20 + 20.00 + 20.10) \text{ cm}^3}{3} \\
 &= 20.10 \text{ cm}^3.
 \end{aligned}$$

01 Mark

Summary

20 cm³ of a solution A₁ required 20.10 cm³ of a solution A₂ for complete reaction. 01 Mark

Part II

Table of results.

Experiment	Pilot	1	2	3
Final volume (cm ³)	23.90	23.60	23.60	23.70
Initial volume (cm ³)	0.00	0.00	0.00	0.00
Volume used (cm ³)	23.90	23.60	23.60	23.70

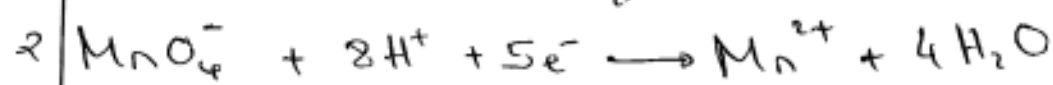
Filling the table = (01)
 Decimal places = (01)
 Accurate volume = (01)
 Total = (03)

$$\begin{aligned}
 \text{Average volume used} &= \frac{V_1 + V_2 + V_3}{3} \\
 &= \frac{(23.60 + 23.60 + 23.70) \text{ cm}^3}{3} \\
 &= \frac{70.9}{3} \\
 &= \underline{23.63 \text{ cm}^3} \quad \text{0/Mark}
 \end{aligned}$$

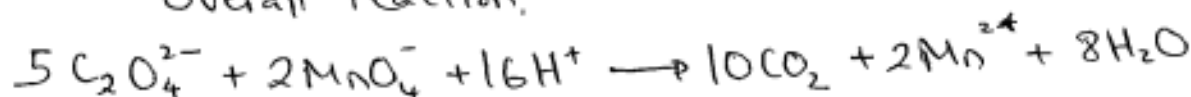
a) Calculate the;

1) Molarity of potassium permanganate.

Considering a balanced chemical equation.



Overall reaction.



$$n_o = 5$$

$$n_p = 2.$$

Finding the molarity of sodium oxalate;

$$\text{Conc. in g/dm}^3 = \frac{m}{V} = \frac{3.35}{0.5 \text{ dm}^3} = 6.7 \text{ g/dm}^3 \quad \text{(00/2)}$$

$$\text{Then; Molarity} = \frac{\text{Conc. in g/dm}^3}{\text{Molar mass}} = \frac{6.7 \text{ g/dm}^3}{134 \text{ g/mol}} \quad \text{(0)}$$

$$= \underline{0.05 \text{ M}}.$$

$$\text{From; } \frac{M_p V_p}{n_p} = \frac{M_o V_o}{n_o}.$$

$$M_p = \frac{M_o V_o n_p}{V_p n_o}$$

(00/2)

$$M_p = \frac{0.05M \times 20 \text{ cm}^3 \times 2}{20.10 \times 5}$$

$$M_p = 0.02M$$

(a)

∴ The molarity of potassium permanganate is 0.02M.

i) Concentration of potassium permanganate in g/dm³.

From:

$$\text{Conc. in g/dm}^3 = \text{Molarity} \times \text{Molar mass.}$$

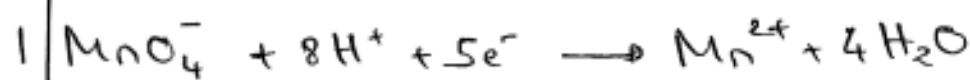
$$= 0.02M \times 158 \text{ g/mol.}$$

$$= \underline{3.16 \text{ g/dm}^3}.$$

(a)

iii) Molarity of iron (ii) salt.

considering a balanced equation,



Overall equation,



(a)

$$n_{\text{Fe}} = 5, n_p = 1.$$

$$\text{From; } \frac{M_{\text{Fe}} V_{\text{Fe}}}{n_{\text{Fe}}} = \frac{M_p V_p}{n_p}$$

(a)

$$M_{\text{Fe}} = \frac{M_p V_p n_{\text{Fe}}}{V_{\text{Fe}} n_p}$$

$$M_{\text{Fe}} = \frac{0.02M \times 20.10 \text{ cm}^3 \times 5}{23.63 \text{ cm}^3}$$

$$M_{\text{Fe}} = 0.085M.$$

(a)

∴ The molarity of iron(ii) salt is 0.085M.

1v) Concentration of anhydrous iron (ii) salt in g/dm^3 .

From;

$$\begin{aligned}\text{Concentration in } \text{g/dm}^3 &= \text{Molarity} \times \text{Molar mass.} \\ &= 0.085 \text{ M} \times 284 \text{ g/mol.} \\ &= \underline{24.14 \text{ g/dm}^3}. \quad (01)\end{aligned}$$

b) Required the value of X .

Given;

$$\text{Conc. in } \text{g/dm}^3 = 33.3 \text{ g/dm}^3.$$

$$\text{Molarity} = 0.085 \text{ M}.$$

$$\text{From: Molar mass} = \frac{\text{Conc. in } \text{g/dm}^3}{\text{Molarity}}$$

$$= \frac{33.3 \text{ g/dm}^3}{0.085 \text{ M}}.$$

$$= 391.76 \text{ g/mol.} \quad \text{--- (01 Mark)}$$

Then;

$$\text{Fe}(\text{SO}_4)_2 \cdot \text{NH}_4 \cdot X \text{H}_2\text{O} = 391.76$$

$$56 + (16 + 64) \times 2 + 14 + 4 + 18x = 391.76$$

$$284 + 18x = 391.76$$

$$\frac{18x}{18} = \frac{107.76}{18}$$

$$x = 6.$$

$\therefore$ The value of X is 6.

(01)

2 Part I.

Initial temperature (T_i) = 32°C .

Final temperature (T_f) = 52°C

$$\Delta T = T_f - T_i$$

$$\Delta T = 52^\circ\text{C} - 32^\circ\text{C}$$

$$\underline{\Delta T = 20^\circ\text{C}.}$$

Part II:

Initial temperature (T_i) = 32°C .

Final temperature (T_f) = 34°C .

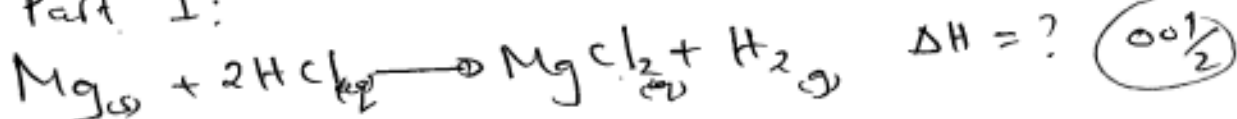
$$\Delta T = T_f - T_i$$

$$\Delta T = 34^\circ\text{C} - 32^\circ\text{C}$$

$$\underline{\Delta T = 2^\circ\text{C}.}$$

a) Required the heat evolved.

Part I:



From: $\Delta H = -m \times c \times \Delta T$.

$$\Delta H = -\rho V c \Delta T$$

$$\Delta H = -1\text{g/cm}^3 \times 50\text{cm}^3 \times 4.2\text{Jg}^{-1}\text{K}^{-1} \times 20^\circ\text{C}$$

$$\Delta H = -4200\text{J} \text{ or } -4.2\text{kJ}.$$

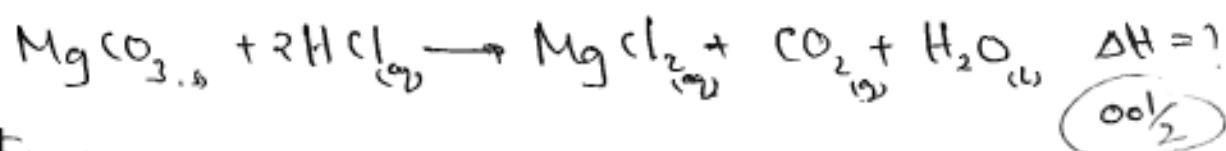
But; $0.2\text{g of Mg} \equiv -4.2\text{kJ}$

$24\text{g of Mg} \equiv ?$

$$-504\text{kJ/mol}.$$

$\therefore$ The heat evolved during the reaction in part I is -504kJ/mol .

Part II.



From; $\Delta H = -mc\Delta T$

$$\Delta H = -\rho V c \Delta T$$

$$\Delta H = -1\text{g/cm}^3 \times 50\text{cm}^3 \times 2^\circ\text{C} \times 4.2\text{J/gK}$$

$$\Delta H = -420\text{J} \text{ or } -0.42\text{kJ}$$

But;

$$1\text{g of MgCO}_3 \rightarrow -0.42\text{kJ}$$

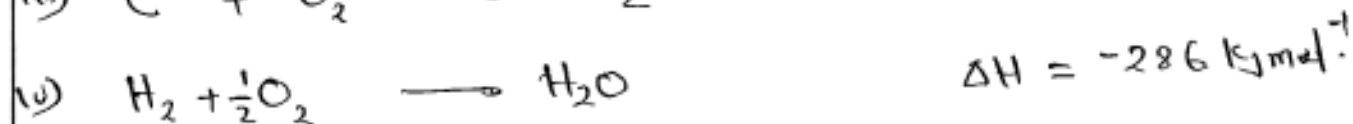
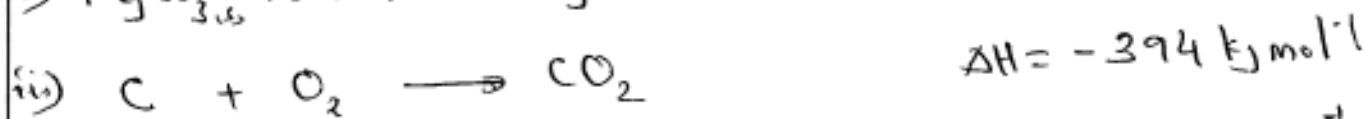
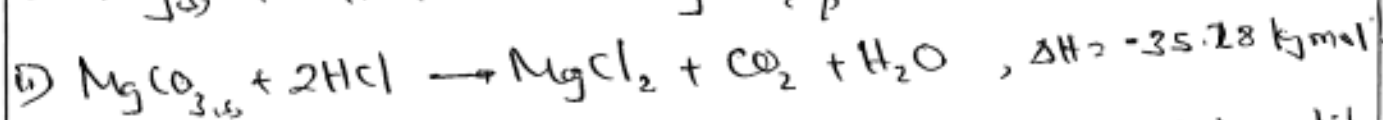
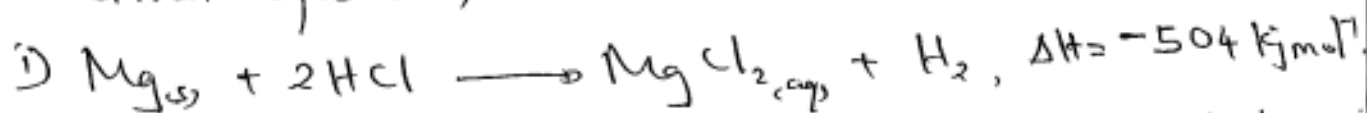
$$84\text{g (1mol) of MgCO}_3 \rightarrow ?$$

$$\Delta H = -35.28\text{kJ/mol}$$

∴ The evolved during the reaction in part II is -35.28kJ/mol .

b) Required the enthalpy of MgCO_3 .

Given equation;

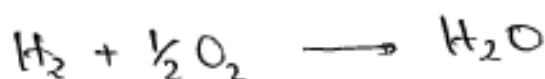
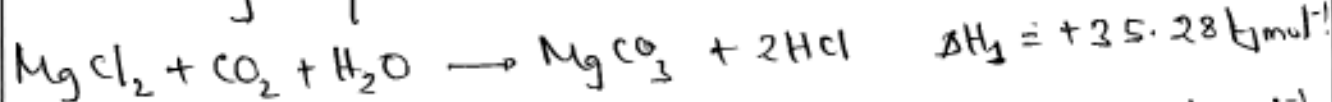


Required equation;



00½@ = 02½ Marks

Reversing equation (iv)



(03)

$$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$$

(001/2)

$$\Delta H_f = +35.28 + -504 + -394 + -286 \text{ kJ mol}^{-1}$$

$$\Delta H_f = -1148.72 \text{ kJ mol}^{-1}$$

∴ The heat of formation of MgCO_3 was
-1148.72 kJ mol⁻¹.

(01)

Sr	Exp.	Observation c1@	Inference. c0½@
a)	(0½)	Green-yellow colour crystalline form	Fe^{2+} , Ni^{2+} , Cr^{3+} , Cu^{2+} , Fe^{3+} CrO_4^{2-} may be present. NO_3^- , SO_4^{2-} , Cl^- , $\text{C}_2\text{O}_4^{2-}$, CrO_4^{2-} , NO_2^- , CH_3COO^- $\text{Cr}_2\text{O}_7^{2-}$ may be present.
b)	(0½)	Blue-green flame was observed Yellow sparks	Cu^{2+} may be present. Fe^{2+} , Fe^{3+} may be present.
c)	(0½)	Black/Reddish brown residue was observed.	Cu^{2+} , Fe^{2+} , Fe^{3+} may be present.
d)	(0½)	Colourless gas with irritating smell evolved, which turns moist litmus paper from blue to red and form dense white fumes with ammonia gas.	Cl^- may be present.
e)	(0½)	Dirty green precipitate insoluble in excess turns brown on exposure to air.	Fe^{2+} may be present.
f)	(0½)	i) Dark blue precipitate was formed.	Fe^{2+} confirmed.
	(0½)	ii) Blue precipitate soluble in excess ammonia forming a deep blue solution.	Cu^{2+} confirmed.
g)	(0½)	White precipitate soluble was formed.	Cl^- confirmed.

Conclusion.

- i) The cations in Sample M are Fe^{2+} and Cu^{2+} . (01)
- ii) The anion in Sample M is Cl^- (01)
- iii) The molecular formula M is FeCl_2 and CuCl_2 . (01)

Total = 15 Marks

